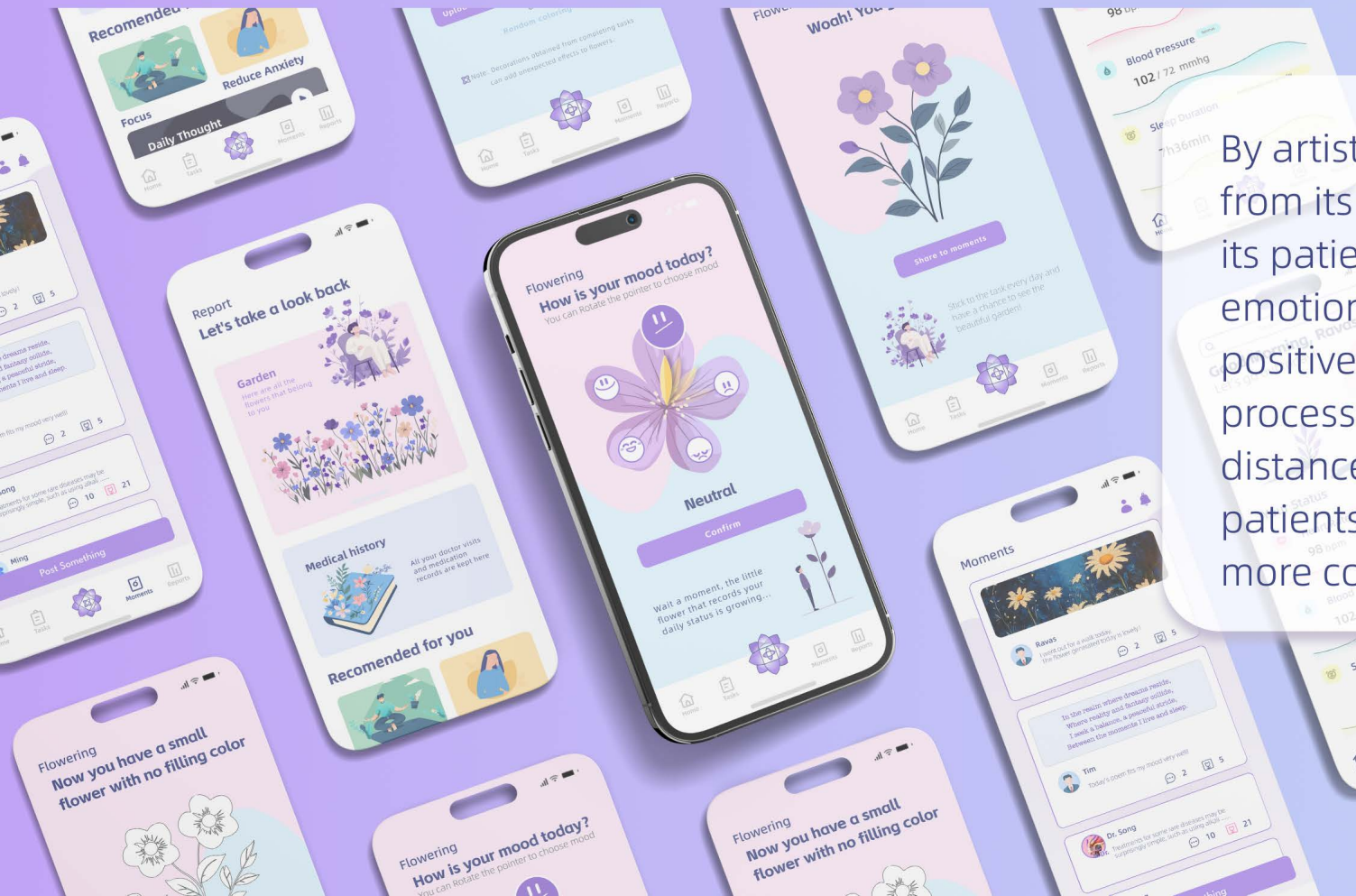


FLOWERING

Service design in the stage of medical prognosis

Project member: Zhang Yufei, Zhu Yichao, Song Haiyu, Xu Xizjie, Shen Nuoqing

Institution: ShanghaiTech University



By artistically collecting physical data from its users, Flowering encourages its patients to pay attention to their emotional state and maintain a positive attitude. In addition, this process can effectively shorten the distance between doctors and patients, making medical services more convenient and close.

BACKGROUND

In daily life, many people who suffer from serious illness, despite receiving physical treatment, do not receive continuous prognostic tracking and emotional support. Because patients often face a sense of disability caused by disease, physical degradation, and other complications after surgery, which leads to increasing unhappiness of patients.

RESEARCH

Health archives, health knowledge services, and health data analysis are the most covered parts of domestic health and medical services, but such services are still difficult to gain popular support because of the complexity of file information, the authenticity and authority of knowledge are not guaranteed, and the need to strengthen privacy protection.

Through interviews with patients, their families and medical staff, we considered using artificial intelligence to solve the problems of patient information collation and knowledge base building, while using artistic healing methods to help patients relieve feelings of disability and anxiety.





 Age: 32
 Occupation: Public hospital doctor

"Life is such a big thing, no one can jump to conclusions."

Too busy to have a hobby

Dislike phone calls

Demand and extent * The filled grid is very urgently needed

It is hoped that the prognostic status data of patients can be easily collected

Clear and reasonable division of responsibilities to reduce disputes



 Age: 54
 Occupation: Used to work for xx company, resigned after illness

"I have become a burden to my family..."

Cancer patient

Lose heart

Demand and extent

I hope the disease will be cured and my life will resume

Reduce anxiety



 Age: 27
 Occupation: Manager of x Company

"How do you cheer him (patient) up?"

Migrant worker

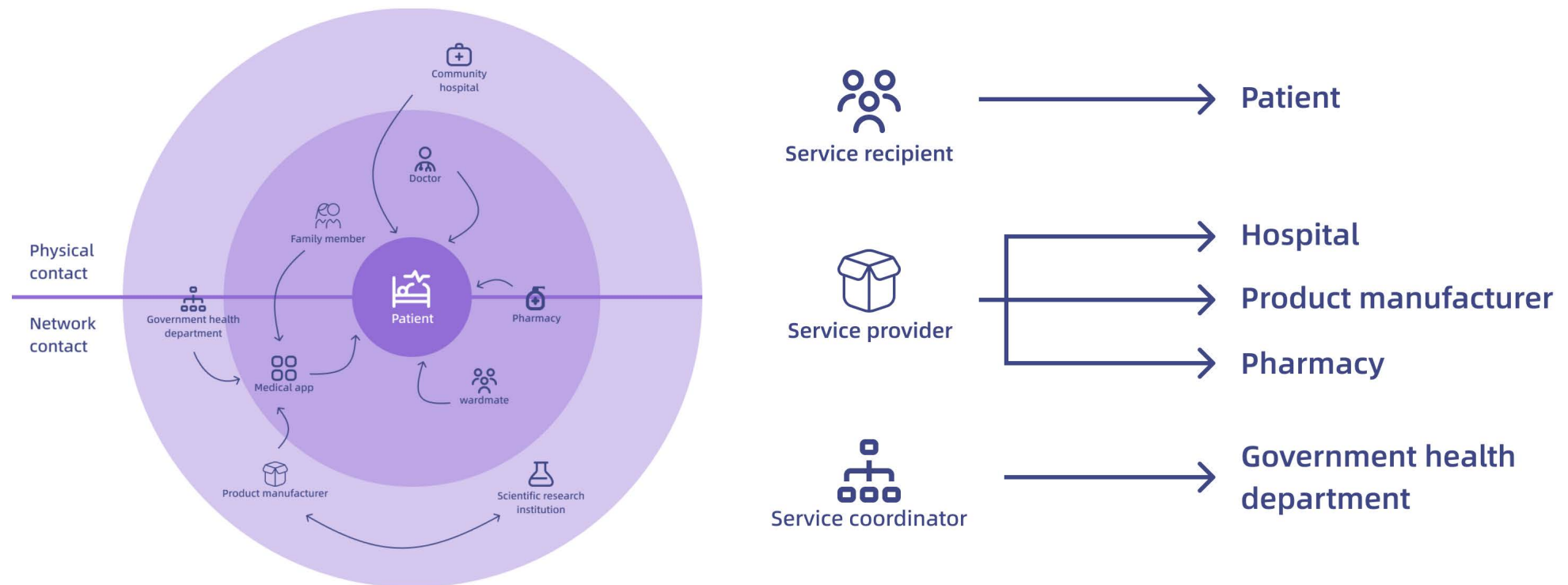
Worry for the patient

Demand and extent

Hope to find a reliable treatment and possible course of the disease

The hope is to relieve anxiety and other stress in patients

STAKEHOLDER MAP



In the system supported by FLOWERING, patients' medical records will be entered into a national network database and be strictly safeguarded, with only authorized doctors and patients having access. Through the mobile application, the daily health data and status of patients will be collected in real time, and the system will promptly alert the responsible medical institutions as soon as any anomalies are detected. Additionally, the app also offers a social platform where users can not only share a positive attitude towards life but also directly book a doctor's outpatient service.

The system is also furnished with a knowledge base driven by artificial intelligence technology, and users can effortlessly retrieve the latest drug information, medication guidelines, and treatment options for diverse symptoms. It can further offer an expected progression trajectory of the disease and assist patients and their families in alleviating anxiety. All the contents of the knowledge base are reviewed by professionals and supervised by regulatory bodies to guarantee the accuracy and reliability of the information.

Health management function

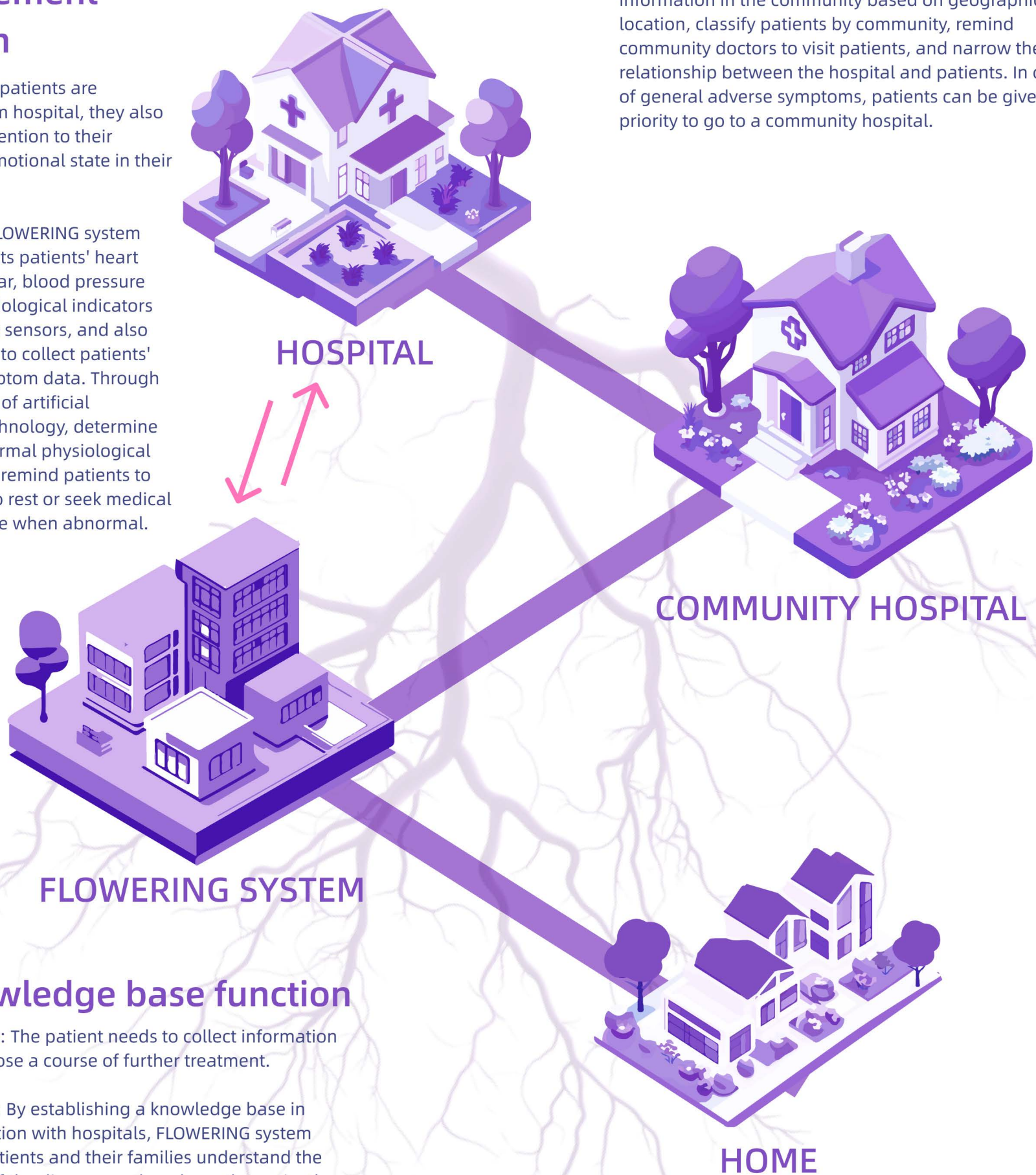
Scenario: After patients are discharged from hospital, they also need to pay attention to their physical and emotional state in their daily life.

Services: The FLOWERING system passively collects patients' heart rate, blood sugar, blood pressure and other physiological indicators through mobile sensors, and also provides a way to collect patients' mood and symptom data. Through the application of artificial intelligence technology, determine the range of normal physiological indicators, and remind patients to pay attention to rest or seek medical attention in time when abnormal.

Medical follow-up function

Scenario: Regular offline contact with the patient and collection of opinions and suggestions after the patient's discharge from hospital.

Services: The FLOWERING system can record patient information in the community based on geographical location, classify patients by community, remind community doctors to visit patients, and narrow the relationship between the hospital and patients. In case of general adverse symptoms, patients can be given priority to go to a community hospital.



Knowledge base function

Scenario: The patient needs to collect information and choose a course of further treatment.

Services: By establishing a knowledge base in cooperation with hospitals, FLOWERING system helps patients and their families understand the course of the disease, and analyzes the patient's situation through artificial intelligence technology, and puts forward possible treatment suggestions. The content in the knowledge base is shared by patients themselves or uploaded by the hospital for desensitization, which is reviewed by the hospital or the government health department.

Record management function

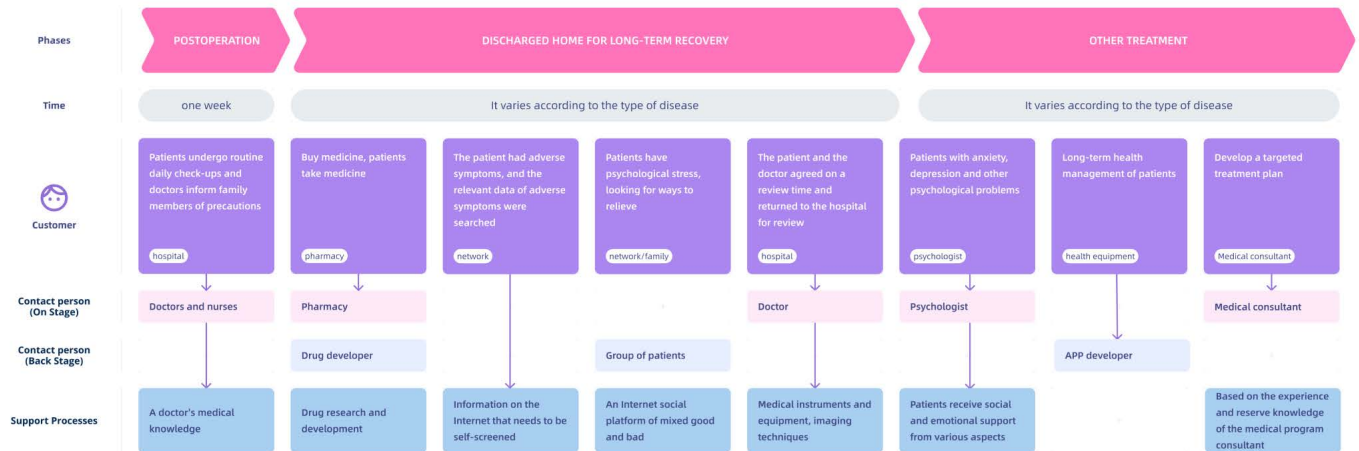
Scenario: When the patient goes to the doctor for follow-up, he or she wants to recall the previous medical experience and information.

Services: The FLOWERING system provides a private and reliable database, all the patient's medical records will be organized and saved, and can be consulted when needed. When a patient receives treatment in a public hospital, the public hospital shall be identified as the responsible hospital for the patient, and the attending physician shall be the responsible physician. Only the responsible doctor and the patient have permission to view the patient's medical records, medication records and other information. When the responsible hospital or doctor changes, the authority will be transferred accordingly.

SERVICE BLUEPRINT

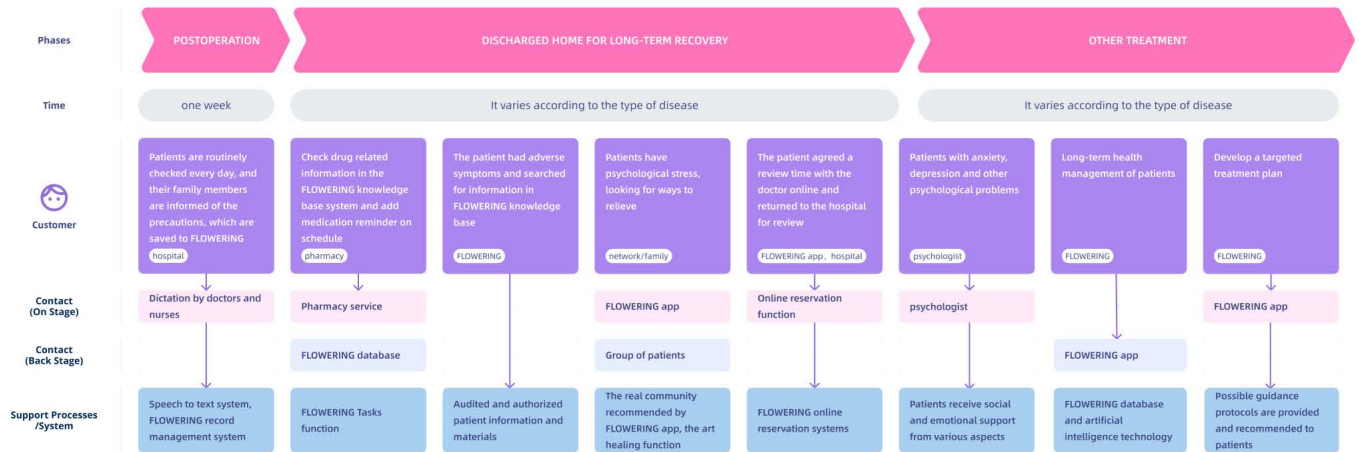
without FLOWERING

1. Post-operative precautions should be recorded and followed by patients and their families
2. The purchase and administration of drugs require the patient to remember
3. No trusted free platform can be found when searching for relevant information
4. Most of the patient communities that are joined on the Internet are set up to sell goods
5. Long-term health management apps are mostly developed for healthy people
6. Medical consultants are hard to find and expensive



with FLOWERING

1. Post-operative precautions are recorded and reminded by the FLOWERING app
2. The app reminds patients to buy and take their medication
3. Search the audited and authorized knowledge base to find relevant information
4. FLOWERING recommends verified communities, free of misinformation and advertising
5. FLOWERING app is designed and developed specifically for people with a recovering prognosis
6. FLOWERING app provides an online, AI-powered plan making solution. The plan making process can be found in the source



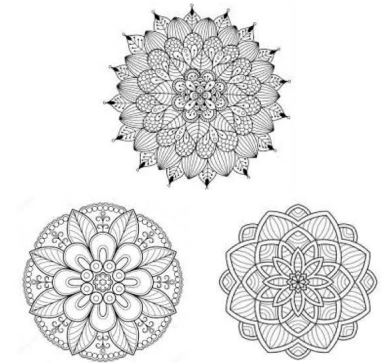
LOGO design



Source of inspiration: **Mandala therapy**

Explanation: Mandala therapy is a type of psychotherapy in which mandala is a Sanskrit word meaning "circle" or "center".
Mandala painting therapy believes that by drawing patterns with specific geometric shapes and symmetries, individuals can help themselves to self-expression and self-discovery, and then achieve the effect of psychological therapy.

Flowering collects user data to generate daily flowers, thereby visualizing the daily state of patients while encouraging them to respond positively to treatment.



Icon - Navigation bar



Home



Tasks



Moments



Reports



Home



Tasks



Moments

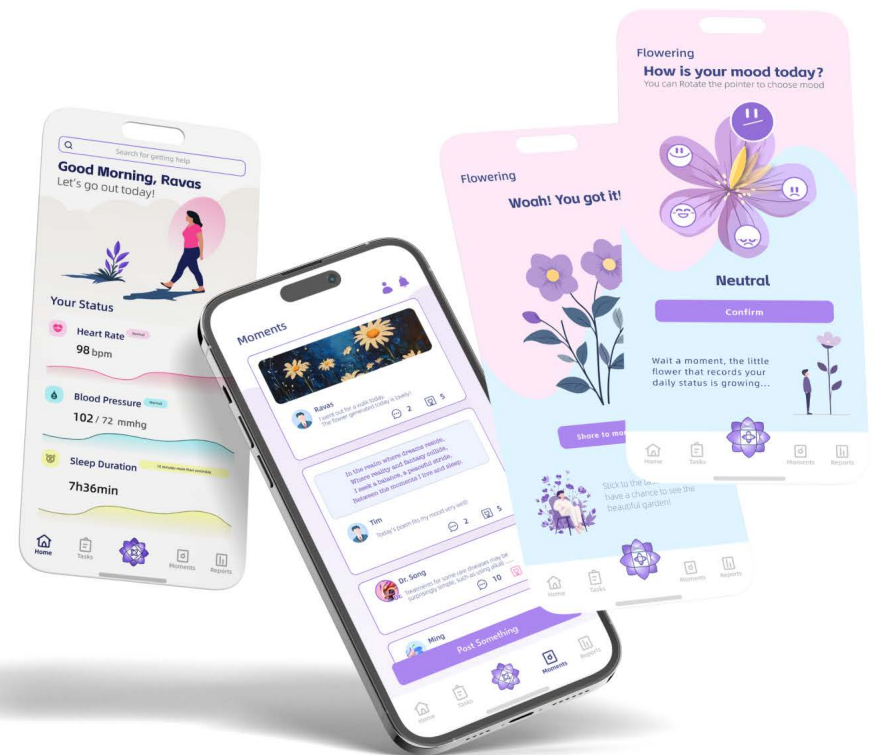
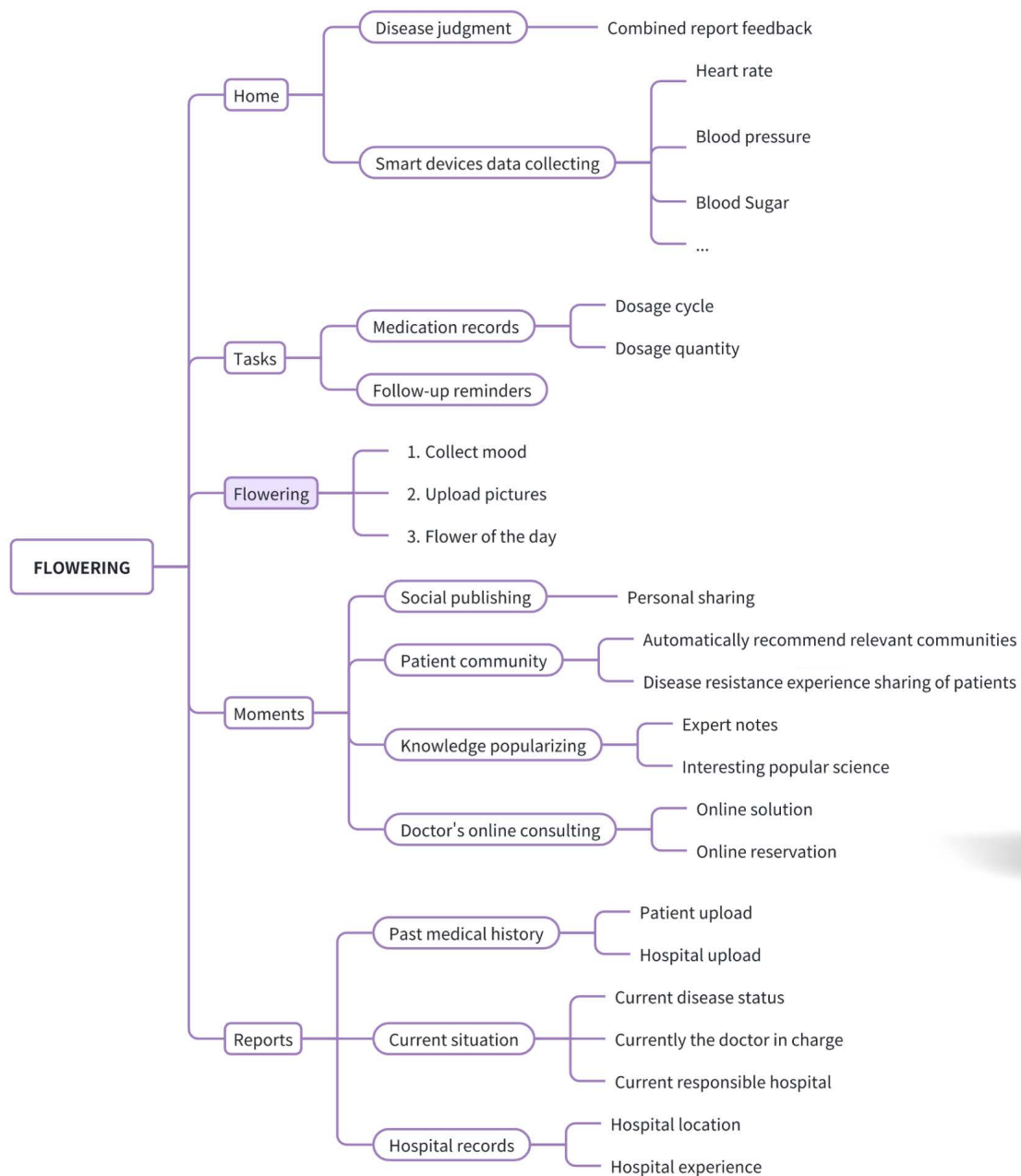


Reports

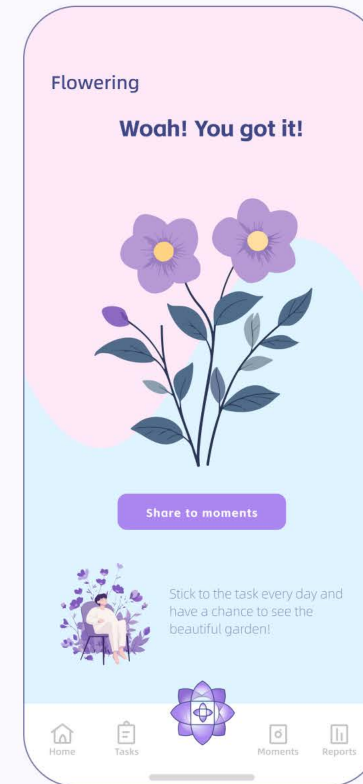
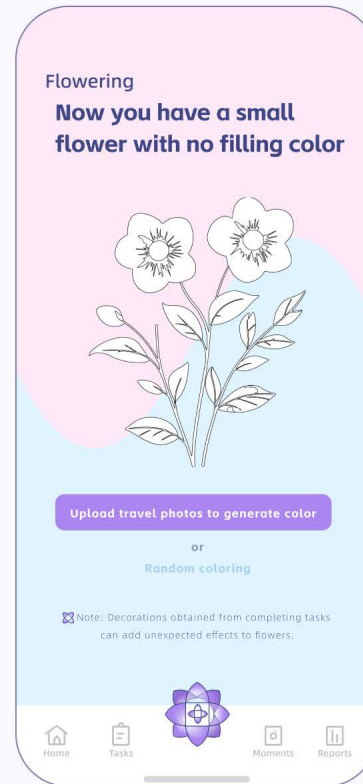
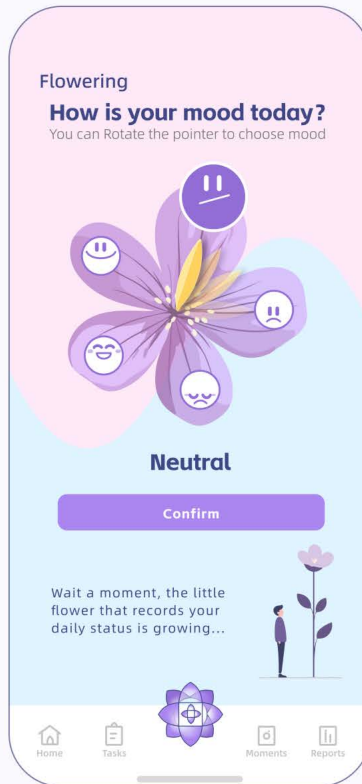
Icon - Mood collection



Functional architecture



Main function display - FLOWERING



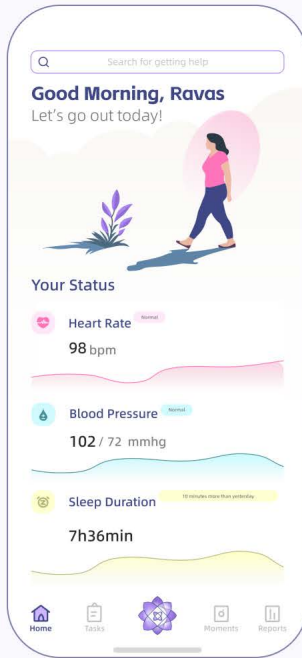
Turn the golden petal needle to choose your mood Today

The AI combines the body data collected by the device to generate a floral thread

Upload photos or random colors

AI takes colors to generate flower colors

Generated flowers are included in Report-Garden
Optionally share to Moments



Home

© Body indicators collected in real time through the wearable device are displayed on the home page, and users can freely manage the device connection.

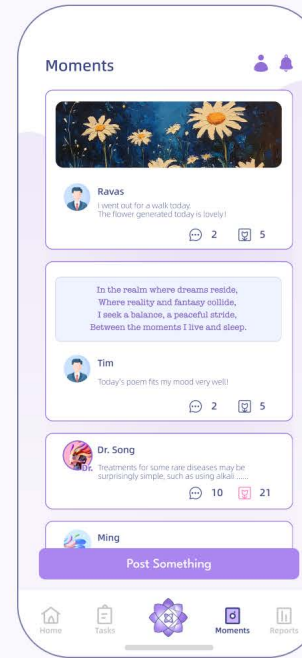
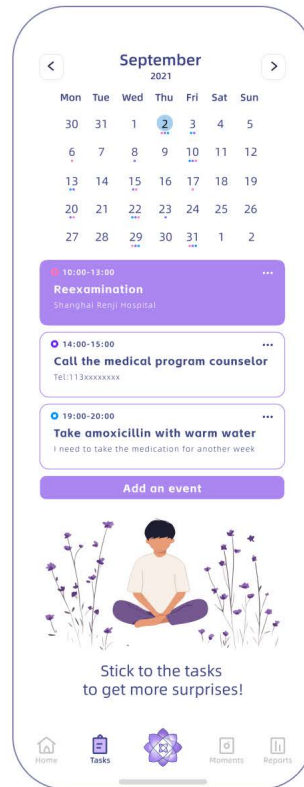
© If an indicator of the body is Abnormal, the status identifier will be displayed as "ABNORMAL"; If the indicator is abnormal for a long time, the APP will push medical suggestions and reminders.

© The search bar at the top of the home page allows users to find videos on how to relieve adverse symptoms, information on medicines and how to buy them. All links provided are verified by the official authorities.

Tasks

© The to-do list will display the to-do agenda for each day of the month, and the type of agenda is distinguished by small colored circles.

© Users can manually enter the time, location and related schedule information, or they can sync their to-do list from the hospital, clinic or FLOWERING's online booking function.



Moments

© Users can share their daily experiences in the community, communicate with certified doctors or submit an online appointment.

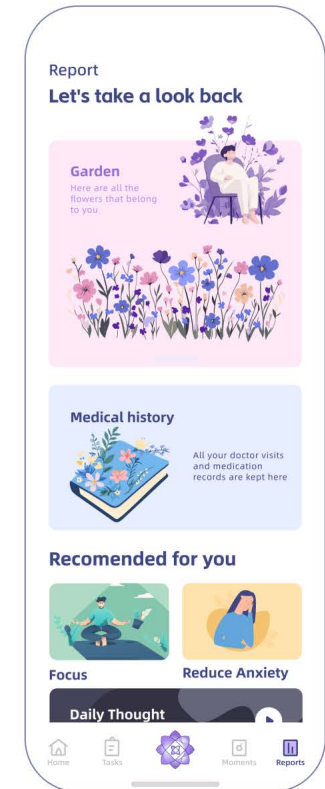
© FLOWERING encourages users to communicate online, and also encourages users to share quality content and spread positive energy.

Reports

© Symptom data collected by the app through the smart device will be stored in the Garden's raw data and can be viewed.

© Flowers generated in FLOWERING will be collected in the Garden.

© FLOWERING will provide life tips based on daily reports.



ADDITIONAL STATEMENT

This project is a course project of User experience and Innovative design course in 2024 Spring. I participated in the early online survey, competitive product survey, interview of medical workers, service blueprint drawing, high-fidelity prototype drawing and other work. Project documents are included in:

<https://vu19zbapr9.feishu.cn/docx/BzlkdfGboopDhxJwUdcFp6Lnhf>

(Part of the design document is different from the document because I made design improvements after the classes.)

At present, China's medical system does have some problems such as excessive pressure on doctors, strained relationship between doctors and patients, and poor patient experience. This is partly because China has a large population base and relatively scarce medical resources, and partly because the existing system does not make full use of the advantages of informatization and technology. I wanted to use interaction design to really revolutionize the system, get rid of formalism, and get things done more efficiently.